

Practice Session 5

In this practice session we will introduce the bootstrap distribution and the bootstrap confidence interval. You may use the functions `sample()` and `do_it()` to generate the bootstrap distribution, and `cnorm()` to find the critical value corresponding to a specific **confidence level**.

Part 1: Confidence interval concept

Practice 1: True or False — Confidence interval interpretation

A catalog sales company promises to deliver orders placed on the Internet within 3 days. Follow-up calls to a few randomly selected customers show that a 95% confidence interval for the proportion of all orders that arrive on time is $85\% \pm 5\%$.

1. Between 80% and 90% of all orders arrive on time.
2. 95% of all random samples of customers will show that 85% of orders arrive on time.
3. The interval between 80% and 90% gives a plausible range of values for where the true population parameter lies since 95% of intervals created will contain the population proportion.
4. For a given sample size, higher confidence means a larger margin of error.
5. For a specified confidence level, smaller samples provide smaller margins of error.

Part 2: Construct Bootstrap Distribution

Here's the clever idea: We don't have the population, but we have a sample. Probably the sample is similar to the population in many ways. So let's sample from our sample. We'll call it **bootstrapping**. We want samples **the same size** as our original sample, so we will need to **sample with replacement**. This means that we may pick some members of the sample more than once and others not at all. We'll replicate this many times.

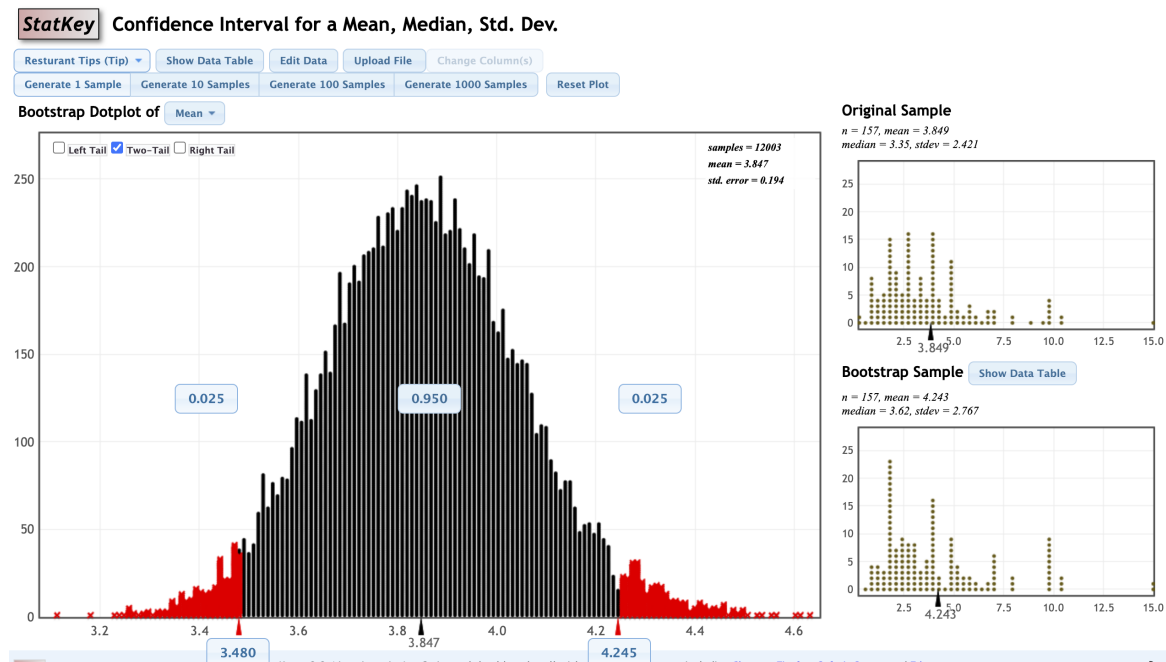
Generating a Bootstrap Distribution:

- Generate bootstrap samples by **sampling with replacement** from the original sample, using the **same sample size**.
- Compute the **statistic of interest** (called a bootstrap statistic), for each of the bootstrap samples.
- Collect the **sample statistics** for many bootstrap samples to create a **bootstrap distribution**.

Example:

Using the StatKey website ([link](https://www.lock5stat.com/StatKey/)), try playing with the creation of the bootstrap distribution from different data. The following picture shows the website and an example of a bootstrap CI for the variable **tip** from the dataset **Restaurant tips**.

https://www.lock5stat.com/StatKey/bootstrap_1_quant/bootstrap_1_quant.html



Practice 2:

The data `ExerciseHours` provide an in-class survey of statistics students asking them about the amount of exercise per week.

1. **First**, create a `histogram` of `Exercise`. What is the `sample size` of the data?

```
library(Lock5Data)
library(SDS1000)
data(ExerciseHours)

# your code here #
```

2. **Second**, create one bootstrap sample from `Exercise`. You might use the function `sample()`.

```
set.seed(5)

# your code here #
```

3. **Third**, you might replicate this one sample 10000 times `with replacement` using the function `do_it()` to create a `bootstrap distribution`

```
# your code here #
```

4. **Fourth**, create a `histogram` for the bootstrap distribution of the variable `Exercise`

```
# your code here #
```

Part 3: Construct Bootstrap Confidence Interval

Reminder:

The steps to Construct Bootstrap Confidence Interval are:

- 1. Compute the statistic from the original sample.
- 2. Create a bootstrap distribution by re-sampling from the sample.
 - Same size samples as the original sample.
 - With replacement.
 - Compute the statistic for each sample.
 - The distribution of these statistics is the bootstrap distribution.
- 3. Estimate the standard error SE by computing the standard deviation of the bootstrap distribution.
- 4. Create the 95% CI using the formula: $statistic \pm 2 * SE$. (The 2 here is a convenient rounding of the exact 95% critical value, `qnorm(0.95)` 1.96.)
- 5. Interpret the confidence interval within the context.

Practice 3:

From the `ExerciseHours` bootstrap distribution you created in the previous question, create a 95% CI for the population mean weekly exercise hours.

1. **First**, calculate the mean of `Exercise` from your original sample

```
# your code here #
```

2. **Second**, create the bootstrap distribution of `Exercise`. We did it in the previous question (Practice 2).

```
# your code here #
```

3. **Third**, calculate the standard error of our bootstrap distribution of `Exercise`.

```
# your code here #
```

4. **Fourth**, calculate the 95% CI, which is based on the formula: $statistic \pm 2 * SE$

```
# your code here #
```

5. **Fifth**, interpret the confidence interval within the context.

Part 4: Extra Practice — Create Bootstrap Confidence Interval with Different Confidence Levels

Practice 4:

Note: You might use the function `cnorm()` (after loading the `SDS1000` package) to help you find the critical value corresponding to a confidence level — for example, `cnorm(0.95)` returns the critical value for a 95% confidence interval.

1. Create a 90% CI of the mean **Exercise** in R.

```
# your code here #
```

2. Calculate a 99% confidence interval of the mean **Exercise** using the function `cnorm()`.

```
# your code here #
```

3. Compare the three confidence intervals you have obtained from the three different confidence levels: 90%, 95%, and 99%.